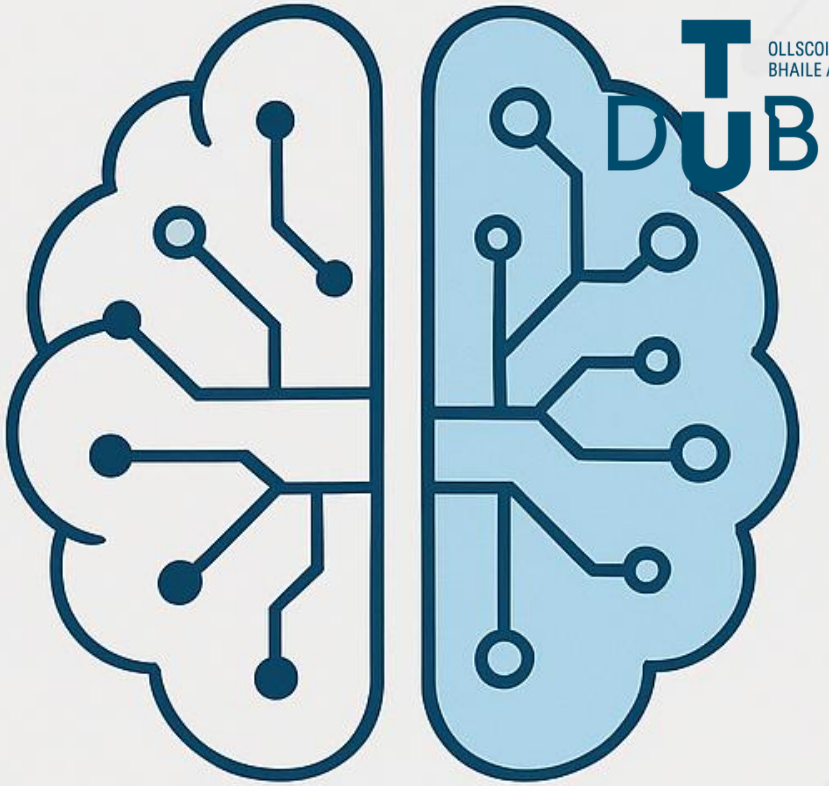
Y2AI PROJECT



OLLSCOIL TEICNEOLAÍOCHTA BHAILE ÁTHA CLIATH
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Artificial Intelligence

Lab 8 – Project Launch

Artificial Intelligence
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COMP 2003 Semester 1 25-26

Lab 8 Y2AI Project Launch

- Review project assignment
- Register yourself/team on Brightspace – Deliverable this week
- Start working !

Introduction

- The purpose of this project is to demonstrate the end-to-end AI development process. You will build and evaluate a small AI system using an open dataset, then use it within a simple demo application
- In this project, you must demonstrate:

- Data set acquisition
- Data cleaning and wrangling
- Feature selection and pre-processing
- Model comparison (at least two algorithms)
- Evaluation (at least two performance measures)
- Development of a simple demo app that uses your model
- Presentation and demonstration of your project
- Specific reflection on ethics and bias related to your project
- Individual written reflection on your contribution and learning

Individual or Group Formation

- This project can be completed individually or as a group of 2–4 members. All group members must come from the same scheduled lab group to facilitate collaboration. You must register your group by the end of Week 8 on Brightspace.
- Group projects must demonstrate a broader and deeper scope: larger or more complex datasets, additional model comparisons (3–4), expanded evaluation (3–4 measures), and a more detailed discussion. Each group must designate one group lead for communication and coordination.

Project Roadmap

Week	Focus	Deliverables
W8	Group or individual registration	Signup on Brightspace
W9	Data wrangling & cleaning	Project Proposal with Dataset & Project Folder on OneDrive
W10	Feature selection & normalization	Interim checkpoint
W11	Model training & evaluation	Draft notebook results
W12	Reflection & submission	Final notebook + reflection
W13	Presentation & Demo	In person presentation/demo

Project Deliverables

- **Project Deliverables**
- End of Week 8 – Group or Individual Project Registration
  End of Week 9 – Project Proposal with Dataset and link to project folder on OneDrive
  End of Week 10 – Upload a short progress report, highlight any issues including non-participation
  End of Week 12 – Final Jupyter notebook and individual reflection
  Week 13 – Presentation and live demo

- Everyone must submit every deliverable including all members of groups

Project Registration – Individual or Group

- You can self-enroll into a project group of 1–4 students in Brightspace under **Module Tools → Groups.** Hit the show all groups button to see available groups, pick **Y2AI Project**
- To work solo, select the next empty group and join it
- To work as a team, all members (up to 4 total) should just join the same initially empty group.
- I will validate joining in LAB and you will also come up with a group name and give it to me.

- Deadline is Friday 7th November 5pm , groups will be locked and you will need to get assistance to update.

Project Proposal Guidelines

- Submit a single-page Word document including:
  - Team Name (if a team)
  - Student Names, Student IDs and email addresses of all participants
  - First name = team lead
- Description of the business problem or project objective outlining the problem that you are solving
- Details of the dataset as per the dataset guidelines
  - Include sizing info e.g. #rows/columns, #images, #docs etc..
  - Confirm that you have checked that you are the only individual/group using this dataset
- Think about the story that you will be able to tell on the back of this project both in the project presentation and in future interviews
- A brief description of how you plan to demo the AI system in action
  - Example: 'User enters a new review , manually classifies it as positive or negative, AI system predicts its judgement – and results are displayed.
  - (A more advanced program might guess review grade from 1 to 5 stars)

Project Infrastructure

- All work must be stored in a shared OneDrive folder accessible to all team members, the lecturer, and lab assistants only.
- Do not share your work outside your team. Backup your folder weekly to avoid data loss.
- You will share a link to this OneDrive location as the W9 deliverable.

Jupyter Notebook Guidelines

- Your notebook must run without errors and include clear markdown documentation. Use markdown headings for each project stage. Avoid embedding commentary as Python comments or print statements. Notebook name must be in the format AIY2X########.ipynb using either your individual or the team leads student number.
- Required markdown cells at the top of your notebook,
  1. Project title
  2. Group/Individual details (names, IDs, emails)
  3. Summary of project proposal
  4. Dataset details
- The notebook must include the following markdown headings
- - Data cleaning and wrangling
  - Feature selection and pre-processing
  - Model and model comparison (at least two algorithms)
  - Evaluation (at least two performance measures)
  - Code that saves your model for use by your demo program
- There must be additional descriptive and analysis commentary in every section.

Dataset Guidelines

- Use a public, ethically sourced dataset with at least 500 rows × 5 features. Include a citation and license. Larger groups must use more complex data. You may not use datasets already covered in lectures or labs. Register your dataset source centrally on Brightspace to ensure uniqueness.
- Its worth spending some time on Kaggle https://www.kaggle.com/datasets, UCI https://archive.ics.uci.edu , the Government of Ireland open data portal https://data.gov.ie.

- Some educational datasets have been collated on Brightspace suitable mainly for individual projects.

- Once you have finalized your project dataset you must register it here.
- You must check to see that the dataset has not been used already. A dataset may only be used by one individual/group.
- Deadline for registering your dataset and submitting the project proposal is 14th November.
- If you are unsure as to the suitability of the dataset please ask for help.
- I reserve the right to reject datasets that are too simple, particularly for groups.

Demo Application

- Your model must be saved and used in a demo program to illustrate functionality.
- The demo may be command-line driven or GUI-based.
- AI tools may assist in demo GUI design, but not in model exploitation or logic development.

Presentation Format (Week 13)

- Presentations occur in lab sessions (schedule TBA). Each member must present a part of the work, and all must attend.
- Presentation durations:
  - Individual: 4 min + 1 min Q&A
  - Pair: 6 min + 2 min Q&A
  - Group of 3–4: 8 min + 2 min Q&A
- Include: dataset & problem overview, model results, demo, bias/limitations reflection, and contributions. Slides are recommended. The demo counts towards both Presentation (10%) and Demo (10%) marks.

Academic Integrity

- All code must be original. External code snippets must be cited. AI tools cannot generate model or logic code. Any dataset related papers or studies must be referenced. All team members share accountability for integrity breaches. Notebooks will be cross-checked and externally checked for academic integrity. Projects are not considered submitted until all deliverables (including personal reflections) are uploaded.

Marking Scheme

- Project total = 100%. Counts as 50% of module grade.
  - Proposal – 5%
  - Dataset sourcing & preparation – 20%
  - Model development – 15%
  - Technical model Evaluation & analysis – 20%
  - Presentation – 10%
  - Demo – 10%
  - Individual reflection – 20%
- Bonus marks may be awarded for complexity, innovation, use of unique datasets (e.g., Irish or sustainability-related), programmatic access to datasets.
- Marks may be reduced for poor markdown formatting, unclear commentary, or lack of justification for design choices.
- Any extensions must be approved prior to the submission deadline with valid reasons. Unapproved late submissions lose up to 10% per working day.

Individual Reflection Guidelines

- Each student must submit an individual reflection, working alone (400–600 words, Word) to Brightspace.

  Suggested structure:
  - Role & Contribution: specific tasks undertaken(if group)
  - Key Learning: technical or conceptual skills developed
  - Challenge: hardest aspect and how it was overcome
  - Ethical/Practical Insight: bias or ethical concerns and mitigation
  - Collaboration Reflection: how work was organized (if group)
  - Future Improvement: what you’d change next time

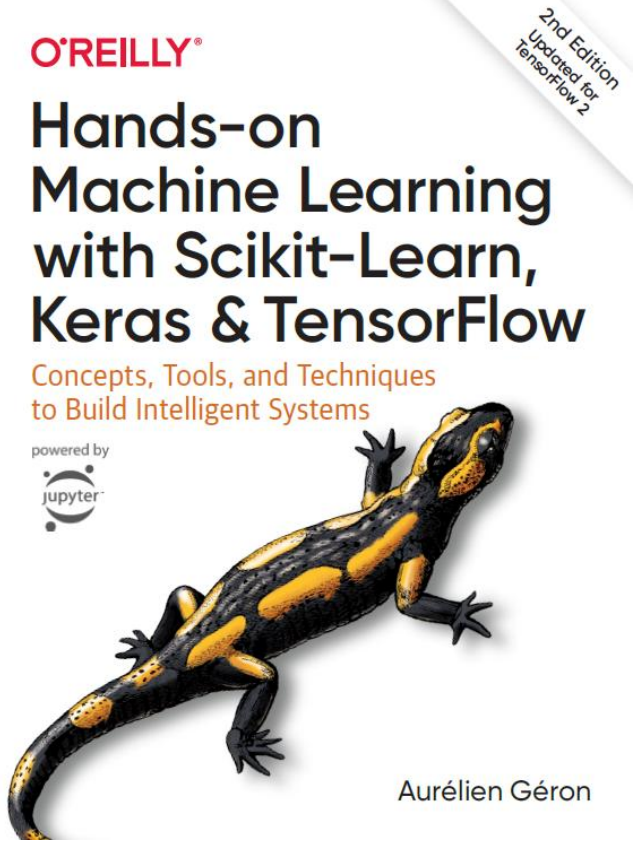
Brightspace Submission Checklist

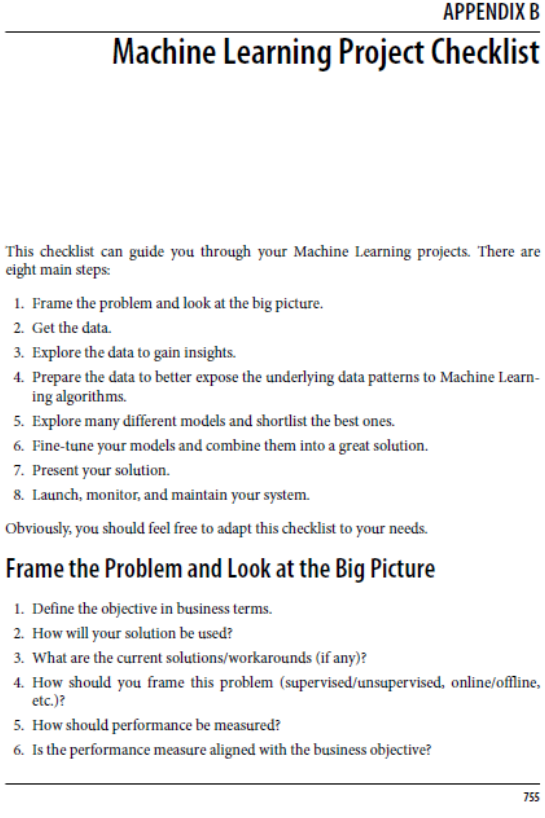
- Whether you do the project as a group or as an individual EVERYONE must submit the following deliverables in Brightspace, as a measure of engagement and progress.

- ☐End of Week 8 – Individual or Group registration
  ☐End of Week 9 – Project Proposal including Dataset
  Link to project folder on OneDrive with preliminary work
  ☐End of Week 10 – Upload a short progress note , highlight any issues or concerns
  ☐End of Week 12 – Final Jupyter notebook and individual reflection
  ☐Week 13 – Presentation and live demo; Must upload the presentation immediately after the pitch

Machine Learning Project Checklist

This checklist is on Brightspace and is a great way to help you organize the project step by step





APPENDIX B

Machine Learning Project Checklist

This checklist can guide you through your Machine Learning projects. There are eight main steps:

1. Frame the problem and look at the big picture.
2. Get the data.
3. Explore the data to gain insights.
4. Prepare the data to better expose the underlying data patterns to Machine Learning algorithms.
5. Explore many different models and shortlist the best ones.
6. Fine-tune your models and combine them into a great solution.
7. Present your solution.
8. Launch, monitor, and maintain your system.

Obviously, you should feel free to adapt this checklist to your needs.

Frame the Problem and Look at the Big Picture

1. Define the objective in business terms.
2. How will your solution be used?
3. What are the current solutions/workarounds (if any)?
4. How should you frame this problem (supervised/unsupervised, online/offline, etc.)?
5. How should performance be measured?
6. Is the performance measure aligned with the business objective?

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